CMS-0042-NC Responses

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Table of Contents

Executive Summary ..... *iii*
1. Patients & Caregivers: From Features to Fundamental Shifts iii
2. Payers: Incentivizing Interoperability Through Market Levers iii
3. Value-Based Care Organizations: Balancing Innovation with Practicality iv
Cross-Cutting Implications v
RFI Responses: *2*
Patients & Caregivers (PC-1 → PC-14) *2*
PC-1 Patient Needs 2
PC-2 Unified Health-Record Access 2
PC-3 Apps Useful to Medicare Beneficiaries 2
PC-4 Missing Features 2
PC-5 Driving Adoption 3
PC-6 Accessibility 3
PC-7 Real-World Evidence 3
PC-8 Data Access & Valuable Sources 3
PC-9 Improving Blue Button 2.0 3
PC-10 TEFCA Impact on Patient Access 4
PC-11 Role of HIEs 4
PC-12 High-Value Operational Use Cases 4
PC-13 Encouraging Information-Blocking Complaints 4
PC-14 Digital Identity Credentials 4
PA-1 TEFCA Limitations for Payers 5
PA-2 Accelerating API Implementation 5
PA-3 Digital Identity for Payers 5
PA-4 Nationwide FHIR Provider Directory Value 5
PA-5 Simplifying Quality Data 5
PA-6 (PA-7 in RFI) Information-Blocking Complaints 6
Value-Based Care Organizations (VB-1 → VB-15) *6*
VB-1 Incentives for ACOs to Use Digital Products 6
VB-2 Integrating AI & Population Health Analytics 6
VB-3 Essential Health IT Capabilities 6
VB-4 Essential Data Types for VBC 6
VB-5 Certification Criteria Support for VBC 6

VB-6 Missing Capabilities in Certification 6
VB-7 Reducing Complexity of Tech Requirements 7
VB-8 Supplementing CEHRT via Other HHS Policies 7
VB-9 Differential Requirements vs Non-APM Orgs 7
VB-10 Flexible CEHRT for Advanced APMs 7
VB-11 Interoperability Challenges in VBC 7
VB-12 Technology Standardization vs Flexibility 7
VB-13 Improving Criteria to Reduce Burden 7
VB-14 Digital Identity & VBC Outcomes 7
VB-15 Nationwide Endpoint Directory Value to VBC 8
Cross-Cutting Enhancements (All Sections) *8*
Public-Private Sandboxes for Rapid Innovation 8
Advanced Consent & Data Intermediation Layer 8
Cognitive Load Index for App Design 8
Appendices *16*
Additions by Section 16
Data Availability & Feedback Loops (PC7–PC10) 16
Payer Innovation (PA-1–PA-7) 16
Value-Based Care (VB-1–VB-15) 16
Futureproofing & Forward-Looking Signals *17*
Emerging Technologies to Watch 17

Executive Summary

1. Patients & Caregivers: From Features to Fundamental Shifts

The collective responses signal a transformation in patient expectations and the health-data ecosystem:

A. **From Passive Portals to Active Navigation:** Patients no longer accept static record repositories; they demand "digital personal assistants" that proactively push alerts, coordinate care across sites, and surface social supports.

> *Implication:* CMS must evolve beyond compliance-driven data sharing toward certifying intelligent care-navigation platforms that combine clinical, financial, and community resource data in real time.

B. **Data Equity as a Trust Builder:** Frustration with siloed records and opaque cost data underlies calls for unified, blockchain-anchored longitudinal records and transparent pricing APIs.

> *Implication:* Mandating richer FHIR endpoints (including imaging and deductible status) and enforcing machine-readable cost data will be essential to rebuild patient trust and dismantle information asymmetries.

C. **Accessibility & Inclusivity as Core Design Principles:** Recommendations for large fonts, offline modes, culturally tailored content, and voice-first interfaces underscore that accessibility is no longer an afterthought but a regulatory imperative.

> *Implication:* CMS certification criteria must codify accessibility standards (e.g., WCAG AA, Project Euphonia support) as non-negotiable prerequisites for any "Digital Seal" of approval.

D. **Real-World Evidence Generation:** Proposals for a "Digital Outcomes Warehouse" and adaptive trials imply a shift from episodic research toward continuous, patient-centric evidence streams.

> *Implication:* CMS can catalyze a real-time learning health system by requiring certified apps to feed de-identified outcomes into a centralized data platform mapped to LOINC/SNOMED.

2. Payers: Incentivizing Interoperability Through Market Levers

Payer-focused responses reveal a clear preference for carrots and sticks to accelerate API adoption:

A. **Regulatory "Nudges" Over Mandates Alone:** Tying MA star bonuses and risk-score uplifts to production-ready APIs suggests payers respond to financial incentives more readily than to unfunded mandates.

> *Implication:* CMS should integrate interoperability performance metrics into existing quality and reimbursement frameworks (e.g., Star Ratings, risk adjustment) rather than invent standalone penalties.

B. **Cost Transparency & Standard Pricing Models:** Calls for a CARIN claims profile in TEFCA and published pricing tiers highlight that unclear fee structures hinder payer participation.

> *Implication:* CMS must publish a transparent, scaled fee schedule for TEFCA participation and build standardized claims APIs to level the playing field for smaller payers.

C. **Digital Identity Embedded in Member Journeys:** Linking premium discounts to Login.gov uptake reflects an expectation that identity proofing becomes a tool for both security and member engagement.

> *Implication:* Embedding digital-credential creation into benefit administration platforms could dramatically reduce help-desk burdens and accelerate secure data exchange.

3. Value-Based Care Organizations: Balancing Innovation with Practicality

Responses from ACOs and other VBC entities emphasize pragmatic flexibility alongside high-ambition technology goals:

A. **Modular Certification Pathways:** The suggestion of an optional "VBC Module" within CEHRT, with lighter requirements for rural providers, acknowledges uneven IT maturity across organizations.

> *Implication:* A tiered certification approach can drive broad baseline interoperability while preserving bandwidth for advanced adopters to innovate in AI, genomics, and real-time analytics.

B. **Incentivizing Digital Therapeutics & AI Governance:** Proposals for shared-savings credits for DTx enrollment and mandatory AI-bias audits indicate that VBC entities view digital interventions as central to their quality and cost goals.

> *Implication:* CMS should incorporate digital-therapy uptake and AI governance plans into MSSP and APM contracts, elevating them from optional pilots to core quality measures.

C. **Data Fusion as the New Frontier:** The emphasis on integrating claims, ADT feeds, SDOH codes, and patient-reported outcomes conveys that longitudinal, multi-modal datasets are the bedrock of population health analytics.

> *Implication:* CMS policies must mandate bulk FHIR exports for both clinical and payer data, coupled with governance frameworks that protect privacy while enabling cross-sector analytics.

D. **Digital Identity for Care Coordination:** Demonstrated reductions in duplicate imaging and login-recovery calls argue that credential portability is essential not only for patients but for cross-organizational care teams.

> *Implication:* Auto-provisioning of high-assurance credentials during enrollment and standardization of credential service providers (CSPs) could become foundational to seamless, multi-site care delivery.

Cross-Cutting Implications

1. **Move from Compliance to Capability:** Stakeholders want agencies to shift from checklist enforcement toward certifying demonstrable outcomes—e.g., reduced readmissions, improved medication reconciliation, and equitable access.
2. **Incentive-Driven Adoption:** Financial and reputational levers (reimbursement uplifts, star-rating bonuses, public uptime rankings) will likely be more effective than pure mandates, especially for payers and VBC organizations.
3. **Equity & Accessibility as Non-Negotiables:** Accessibility features, bias mitigation in AI, and compensation for community contributions must be embedded in policy design from the outset.
4. **Data as a Shared Asset:** Across all sectors, there's a consensus that data—clinical, claims, social—must flow continuously into interoperable platforms to power real-time decision-support, research, and quality improvement.

RFI Responses:

Patients & Caregivers (PC-1 → PC-14)

PC-1 Patient Needs

Patients value digital tools that (1) "push" timely tasks (medication refills, preventive screening-due alerts), (2) integrate wearables/remote sensors to detect early deterioration, and (3) auto-route them to the lowest-cost high-quality site of care. An intelligent care-navigation app should: (a) show real-time deductible status and expected out-of-pocket, (b) offer 24/7 chatbot triage with nurse escalation, (c) surface community SDOH resources (transportation vouchers, food pantries) when social risk is detected, and (d) generate a unified dynamic care plan visible to every clinician in the patient's FHIR-enabled record. If a "digital personal assistant" were available, patients would ask it to: track labs and notify when values drift out of personalized target ranges, schedule imaging when queues are shortest, and coach lifestyle behaviors with gamified nudges proven to raise adherence.1,2

PC-2 Unified Health-Record Access

Only 19 % of Medicare beneficiaries currently report seeing all records in one portal (CMS Pulse 2025). Where implemented—e.g., NZ's "My Health Portal"—unified access cut duplicate imaging 14 % and improved medication reconciliation accuracy 22 %.3 Barriers in the US include (a) payer–provider data siloes; (b) CCD-only export that omits imaging DICOM objects; and (c) fragmented patient identity (multiple log-ins). Blockchain-anchored longitudinal records with W3C verifiable credentials can solve cross-organization trust while preserving privacy.4

PC-3 Apps Useful to Medicare Beneficiaries

High-impact categories: (1) pill-dispenser-integrated adherence platforms (e.g., MedaCube 2.0); (2) hypertension RPM bundles that auto-sync BP to the PCP EHR; (3) cognitive-support apps for mild cognitive impairment; and (4) oncology symptom trackers that trigger nurse call-backs at PRO-CTCAE thresholds. CMS should list "pre-checked" apps in Medicare.gov, analogous to NHS Apps Library, focusing on usability in ≥ 65 age group.5,6

PC-4 Missing Features

Gaps: seamless payer-clinical cost data fusion, AI prognosis explainability, and one-click sharing to family caregivers under granular consent. Apps enabling point-of-care price bundling (episode-level) remain rare due to opaque negotiated-rate files. CMS could expose its Price-Transparency API v2 and require MA plans to supply machine-readable rates.7,8

PC-5 Driving Adoption

A voluntary CMS "Digital Seal" assessing safety, equity impact, and ROI would guide seniors. Criteria: CER showing ≥ 10 % admission reduction or satisfaction gain; WCAG-AA accessibility; and transparent ML bias audit. CMS should tie Chronic-Care Management (CCM) CPT 99490 reimbursement uplift (+$10 PMPM) when a certified app is used ≥ 60 days/quarter and data flow via FHIR g(10).9,10

PC-6 Accessibility

Must-have elements: large-font dynamic text, voice-first commands (leveraging Project Euphonia speech models for dysarthria), offline mode for rural bandwidth, and culturally tailored content.11,12

PC-7 Real-World Evidence

CMS could mandate that certified apps push de-identified FHIR Observation resources to a CMS "Digital Outcomes Warehouse" mapped to LOINC/SNOMED. An adaptive platform trial model (like FDA's Evidence Accelerator) can continuously compare app cohorts vs controls on TCM readmissions.13,14

PC-8 Data Access & Valuable Sources

Readily available data for patients include prescription histories and basic labs via most portals, yet high-value elements—radiology images (DICOM), narrative encounter notes, real-time deductible balances, and home-health episode records—remain elusive. A 2024 multi-center study found only 27 % of portals exposed radiology or pathology images despite FHIR capability.15 Barriers are (a) storage-intensive image formats that vendors hesitate to stream, (b) payer-provider contractual limits on claims feeds, and (c) inconsistent provenance tagging that worries downstream app developers. Claims-plus-clinical fusion produces the greatest incremental utility; one integrated dataset improved diabetes care-gap closure 18 % in a 12-month NY ACO pilot.16 CMS can accelerate access by: requiring MA plans to surface adjudicated claims within 48 h via g(10) endpoints; expanding Consolidated-CCD to include ImagingStudy resources; and funding open-source tooling that re-signs data with digital provenance.

PC-9 Improving Blue Button 2.0

Top additions requested by developers are: (1) Part C encounter data, (2) real-time prior-authorization status, (3) SDOH Z-code flags, and (4) durable-medical-equipment claims. A 2024 usage analysis showed apps that combined Part D fills with Part C encounters reduced 30-day readmissions 9 %.17 Obstacles include lagged data refresh (average 17 days) and lack of OAuth granular-scope tokens. Recommendations: publish a v2 FHIR Subscription API to push updates < 72 h after claim adjudication; align scopes with SMART-on-FHIR granular permissions; and allow verified third-party submission of self-reported vitals back to CMS as Provenance resources.18 Including MedPAC cost-

sharing datasets would let beneficiaries forecast out-of-pocket expenses inside navigation apps.

PC-10 TEFCA Impact on Patient Access

Early TEFCA Qualified Health Information Networks (QHINs) report that patient-initiated queries now account for 8 % of traffic, with median fulfillment < 4 s.19 Success cases include oncology patients downloading outside pathology in rural Oregon via a MyChart-launched TEFCA workflow. Needed improvements: mandate patient-facing FHIR endpoints (not only HL7 v2/IHE), require QHINs to support digital identity federation (Login.gov tokens), and publish quarterly metrics (latency, completeness) to spur competition. Standards working well: FHIR v4.0.1 g(10) plus OAuth 2.0/SMART. Under-used but promising: HL7 Gravity SDOH value sets. Redundancies: DirectTrust pushes duplicating CDA mail should sunset once TEFCA patient pull reaches 90 % coverage.

PC-11 Role of HIEs

High-performing regional HIEs (e.g., Indiana Health Info Exchange) deliver medication fill alerts within 6 h and boast 94 % match accuracy.20 Data value is highest where ADT feeds trigger automated care-manager outreach. To improve, HIEs should expose bulk-FHIR exports for patient-authorized apps and incorporate payer claims to enrich longitudinal histories. CMS could create a performance scorecard tied to Medicaid IAPD funding to propagate best-practice HIE governance.

PC-12 High-Value Operational Use Cases

(1) Binding cost estimates: Real-time prescription and facility fee calculation at scheduling reduced surprise bills by 22 % in a Colorado pilot.21 (2) Third-party scheduling: API-driven appointment booking doubled specialist access speed in Kaiser's Northern CA region. (3) Provider quality matching: Patients selecting surgeons with above-median PROs saw 15 % fewer complications. Near-future: AI-generated "likely wait-time" predictions; long-term but hard: real-time OR slot auctioning across systems.

PC-13 Encouraging Information-Blocking Complaints

Embed a one-click "Report Access Problem" in CMS-endorsed apps that pre-populates ONC's portal with FHIR Provenance. Gamify with small HSA incentive ($5) for verified reports. ONC's 2024 data show complaint volume rose 38 % after vendor-portal links were added.22 More reports surface systemic blockers and accelerate corrective action without harming exchange.

PC-14 Digital Identity Credentials

Challenges: seniors face document-upload friction and distrust biometric scans; providers worry about workflow changes. Benefits: single credential cuts average portal-login

recovery calls 47 %.23 Downsides: potential vendor lock-in and privacy fears if credential providers monetize metadata. Encouraging adoption: CMS could auto-provision Login.gov IAL2 credentials during Medicare enrollment, mail QR code activation kits, and offer a trust-framework that lets CLEAR/ID.me inter-operate. Requiring CSPs for Blue Button access would standardize security while giving patients portability across payers and providers.

Payers (PA-1 → PA-7)

PA-1 TEFCA Limitations for Payers

Limitations: lacks bulk-claims query, no cost-sharing schema, and current fee model unclear. Recommend: add CARIN BB claims profile to TEFCA QTF; publish transparent pricing tiers; and enable SMART Bulk FHIR for risk-adjustment.48,49

PA-2 Accelerating API Implementation

CMS could condition MA Star-Rating bonus on production Patient, Provider, and Payer-to-Payer APIs by 2026. Offer a one-time 0.25 % risk-score bump for early adopters. Publicly rank payers' API uptime.50,51

PA-3 Digital Identity for Payers

Link premium discounts to Login.gov adoption; embed credential creation in member onboarding; auto-federate with provider portals. Savings: $1.20 per member/year help-desk costs.52

PA-4 Nationwide FHIR Provider Directory Value

A master directory with FHIR endpoints + DIDs accelerates prior-auth routing, cutting processing time 35 %. Payers gain accurate network-adequacy analytics. CMS could host and fund maintenance via QHP user fees.53,54

PA-5 Simplifying Quality Data

Payers prefer Bulk FHIR extraction so they compute HEDIS in-house. Encourage via CMS rule allowing substitution of payer-calculated eCQMs if audit passes 95 % concordance. Tech leverage: HL7 Da Vinci DQM.55,56

PA-6 (PA-7 in RFI) Information-Blocking Complaints

Require payers to report quarterly IB incidents; offer civil monetary penalty reduction if self-reported. ONC portal integration via API cuts filing friction.57,58

Value-Based Care Organizations (VB-1 → VB-15)

VB-1 Incentives for ACOs to Use Digital Products

CMS should add a DTx Quality Improvement Credit in MSSP: ACOs showing ≥ 10 % enrollment of high-risk beneficiaries in a CMS-certified app earn 0.5 % shared-savings rate boost. Obstacles: unclear ROI and integration costs. CMS could offer one-time Health IT Modernization loans at 0 % for rural ACOs.97,98

VB-2 Integrating AI & Population Health Analytics

Require Advanced APMs to submit an AI Governance Plan covering bias audits, explainability, and ongoing monitoring; supply standardized SDOH risk-stratification algorithms via open-source.99,100

VB-3 Essential Health IT Capabilities

Mandatory: bidirectional FHIR care-plan, remote device interoperability via IEEE 11073; risk-adjusted dashboards; push-based event notifications. Valuable extras: patient-reported outcome (PROM) ingestion and NLP-driven social-needs extraction.101,102

VB-4 Essential Data Types for VBC

Longitudinal claims, real-time ADT, SDOH Z-codes, PROMIS scores, pharmacy fill gaps, and home IoT vitals.103

VB-5 Certification Criteria Support for VBC

Current CEHRT lacks attribution logic and bulk risk-strat exports. ONC should add "Value-API" criterion exposing patient-attribution roster and network leakage analytics.104,105

VB-6 Missing Capabilities in Certification

Not addressed: closed-loop referral interoperability and payer data synchronization. Add criteria for FHIR Task-based referrals and CARIN BB ingest.106,107

VB-7 Reducing Complexity of Tech Requirements

Create "VBC Module" optional path: providers implement only Risk, Quality, and Care-Coordination APIs; small rurals allowed light version with semi-annual bulk uploads.108

VB-8 Supplementing CEHRT via Other HHS Policies

Extend CoP event-notification requirement to ACO REACH entities; align with TEFCA RLS for routing. Remove duplication by letting ACOs use same API calls for both CoP and MIPS PI submission.109,110

VB-9 Differential Requirements vs Non-APM Orgs

APMs should submit eCQMs via bulk; non-APM can continue QRDA I. Provide reduced eRx reporting for high-performance ACOs.111

VB-10 Flexible CEHRT for Advanced APMs

CMS should define a menu: Oncology APMs must have FHIR Genomics IG; Kidney Care APMs need Device FHIR for home dialysis reporting. Offer rural exceptions: allow certified third-party analytics gateways when EHR not upgradable.112,113

VB-11 Interoperability Challenges in VBC

Patient-matching errors (2 %); delayed claims files; proprietary device data. Solution: national patient identifier via federated CSP; require device vendors to provide IEEE 11073 to FHIR translators.114,115

VB-12 Technology Standardization vs Flexibility

Base stack: FHIR R5, CDS Hooks, Bulk FHIR, SMART. Allow plug-in domains (genomics, imaging) to pick IGs. Maintain innovation by 24-mo IG update cycles.116,117

VB-13 Improving Criteria to Reduce Burden

Collapse PI and Quality submission into single Bulk upload; auto-populate HCC risk; allow ACOs to delegate to QHIN analytics hubs.118,119

VB-14 Digital Identity & VBC Outcomes

Credentialed identity enables cross-org care-plan editing and lowers duplicate imaging. Pilot in Colorado ACO cut readmissions 7 %.120

VB-15 Nationwide Endpoint Directory Value to VBC

Directory accelerates data pulls for benchmarking and leakage capture; key fields: FHIR endpoint, network contract ID, quality-star rating.121,122

Cross-Cutting Enhancements (All Sections)

Public-Private Sandboxes for Rapid Innovation

Establish CMS-led regulatory sandboxes with NIH, FDA, and ONC collaboration. Example: Developers test RPM/AI tools with real-time claims and clinical data, temporary billing code waivers.

Advanced Consent & Data Intermediation Layer

Fund an open-source FHIR Consent Broker Layer aligned with Kantara and UMA 2.0 frameworks for granular, revocable permissions and multi-party delegation.

Cognitive Load Index for App Design

Introduce a "Cognitive Load Index" for certified apps, weighted in Digital Seal criteria, validated against NIH/NIA age-appropriate UX standards.

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Appendices

Additions by Section

Patients & Caregivers (PC1–PC4)

PC1 Digital Nudging Framework
Pilot gamified behavioral-economic nudges under the Medicare Innovation Center for adherence improvements.

PC2 AI-Enhanced Record Summarization
Support for LLM-based patient record summarizers with transparency standards for hallucination detection.

PC4 Dynamic Episode Bundling Explorer
Fund a "Bundled Price Explorer" UI leveraging real-time payer APIs and claims data to show bundled care cost estimates.

Data Availability & Feedback Loops (PC7–PC10)

PC7 Embedded Pragmatic Trials via Claims + App Data
Enable pragmatic trials using combined claims and app telemetry data, streamlining registration and reducing IRB overhead.

PC9 Developer-to-CMS Secure Messaging Channel
Introduce secure FHIR-based "issue ticket" communication channels between certified app developers and CMS/ONC for rapid bug reporting.

PC10 QHIN Scorecard for Patient-Facing Metrics
Publish patient-centric TEFCA QHIN performance metrics quarterly (query completion rates, latency, SDOH support).

Payer Innovation (PA-1–PA-7)

PA-1 Integration with CMS Enterprise Identity
Encourage integration of Login.gov identity services within payer portals, aligning member authentication.

PA-4 Smart Prior-Auth Prediction API
Promote a machine learning-enabled prior-authorization prediction API based on historical denial patterns.

PA-6 Payer Penalty Mitigation Credit
Offer mitigation credits for self-reporting of information blocking accompanied by remedial action plans.

Value-Based Care (VB-1–VB-15)

VB-1 App Interoperability Bonus for MSSP ACOs
Integrate metrics on CMS-certified digital tool adoption into MSSP scoring, linked to patient activation measures.

VB-3 Closed-Loop Care Goal Tracking
Require CEHRT vendors to support SMART-on-FHIR apps for real-time patient visualization of health goal progress.

VB-5/VB-10 Value-API Developer Toolkit
Provide ONC-funded SDKs for standardized value-based care modules (attribution, HCC tagging, leakage analytics) for smaller ACOs.

VB-14 National Digital Identity Verifier Registry
Establish CMS/ONC-backed registry of digital identity providers to ensure interoperability, security, and reduce vendor lock-in.

Futureproofing & Forward-Looking Signals

Emerging Technologies to Watch

- AI for patient summarization (FHIR-compatible GPT tools)
- Synthetic health data for robust app testing
- Digital twins for chronic disease management
- Zero-knowledge proofs for consent verification
- QR-code-based health credentials aligned with WHO smart health cards